Logarithmic laws

1 Write each of the following expressions as a single logarithmic term:

a $\log_{10} 5 + \log_{10} 4$

b $\log_{10} 18 - \log_{10} 3$

c $\log_{10} 7 - \log_{10} 28$

d $\log_5 11 + \log_5 2 + \log_5 9$

e $\log_{10} 5 + \log_{10} 7 - \log_{10} 3$

f $\log_7 12 - (\log_7 2 + \log_7 3)$

g $3\left(\log_{10} 9 + \log_{10} 2\right)$

h $3\left(\log_{10} 6 - \log_{10} 2\right)$

i $2\log_5 22 - 2\log_5 11$

j $5\log_{10} 6 + 5\log_{10} 3$

k $3 + \log_4 7$

l $\log_3 (5x) + \log_3 (2y)$

2 Evaluate the following logarithmic expressions:

a $\log_{10} 2 + \log_{10} 5$

b $\log_6 12 + \log_6 18$

c $\log_2 72 - \log_2 9$

d $\log_4 8 + \log_4 2$

e $\log_3 2 - \log_3 18$

f $\log_6 12 + \log_6 15 - \log_6 5$

3 Rewrite the following expressions without any powers or surds:

a $\log_4 \left(x^7\right)$

b $\log\left((x+6)^5\right)$

c $\log\left((3x+7)^{-1}\right)$

d $\log\left(x^{\frac{2}{5}}\right)$

e $\log_6 y^5$

f $\log_6 \sqrt{w}$

g $\log_{10} \sqrt{10}$

h $\log_{10} 10^{\frac{5}{4}}$

i $\log_{10} \left(10^a\right)$

j $\log_6 (36)^5$

k $\log_5 \sqrt{x^3}$

l $\log_5 125^{\frac{5}{4}}$

4 Rewrite the following as the sum or difference of logarithms without any powers or surds:

a $\log_9 (uv)$

b $\log_5 \left(\dfrac{9}{7}\right)$

c $\log_2 (5x)$

d $\log_{10} \left(\dfrac{2}{x}\right)$

e $\log\left(m^2\right)$

f $\log\left((3x)^5\right)$

g $\log\left(\dfrac{pq}{r}\right)$

h $\log\left(\dfrac{1}{xy}\right)$

i $\log\left((5x)^{-7}\right)$

j $\log\left(5x^{\frac{2}{3}}\right)$

k $\log\left((14x)^{\frac{1}{3}}\right)$

l $\log\left(\sqrt{\frac{c^8}{d}}\right)$

5 Write each of the following as a single logarithm or integer:

a $5\log x^3 - 4\log x^2$

b $5\log x + 3\log y$

c $8\log x - \dfrac{1}{3}\log y$

d $7\log x - \log\left(\dfrac{1}{x}\right) - \log y$

e $7\log_{10} 5 - 21\log_{10} 25$

f $5\log_{10} 8 - 3\log_{10} 4$

g $2\log_6 3 + \dfrac{1}{3}\log_6 64$

h $\log_2 36 - 2\log_2 3$

Write $\log\left(\frac{2u}{3v}\right)$ in terms of $\log 2$, $\log u$, $\log$ [illegible] $\log v$.

7 Rewrite the expression $\log x^2 + \log x^3$ in the form $k \log x$.

8 Rewrite the following in terms of $\log u$ and $\log v$ without any powers or surds:

a $\log\left(u^3 v^5\right)$

b $\log\left(\frac{\sqrt[3]{v}}{\sqrt{u}}\right)$

9 Simplify each of the following expressions:

a $\frac{5 \log m^2}{6 \log \sqrt[3]{m}}$

b $\frac{\log a^8}{\log a^4}$

c $\frac{\log a^3}{\log \sqrt[3]{a}}$

d $\frac{\log\left(\frac{1}{x^4}\right)}{\log x}$

e $\frac{\log_{10} 4}{\log_{10} 2}$

f $\frac{\log_4 125}{\log_4 5}$

g $\log_{10} 10 + \frac{\log_{10}\left(15^{20}\right)}{\log_{10}\left(15^5\right)}$

h $\frac{8 \log_{10}\left(\sqrt{10}\right)}{\log_{10}(100)}$

10 If $\log_a 3 = 1.16$ and $\log_a 2 = 0.73$, find the value of $\log_a \sqrt{54}$.

11 Using the following rounded values, evaluate the expressions below correct to three decimal places:

- $\log_{10} 7 = 0.845$
- $\log_{10} 2 = 0.301$
- $\log_{10} 3 = 0.477$
- $\log_{10} 25 = 1.398$
- $\log_{10} 5 = 0.699$

a $\log_{10} 28$

b $\log_{10}\left(\frac{5}{3}\right)$

c $\log_{10} 49 + \log_{10} 27$

d $\log_{10} 49 - \log_{10} 125$

12 Using the rounded values $\log_x 3 = 0.62$ and $\log_x 4 = 0.78$, find the value of each of the following expressions:

a $\log_x 9$

b $\log_x \sqrt{3}$

c $\log_x 4x$

d $\log_x \frac{1}{3}$

e $\log_x 36$

13 Given that $\log_b x = 2.6$ and $\log_b y = 4.2$, find the value of each of the following expressions:

a $\log_b x^3$

b $\log_b \sqrt[3]{y}$

c $\log_b \left(x^2 \sqrt{y}\right)$

d $\log_b \left(\frac{b}{x}\right)$

14 Rewrite $\log_3 20$ in terms of base 4 logarithms.

15 Rewrite the following in terms of base 10 logarithms:

a $\log_4 16$

b $\log_3 0.9$

c $\log_3 \sqrt{5}$

16 Consider the following logarithmic expressions:

i Rewrite the expression in terms of base 10 logarithms.

ii Hence, evaluate each to two decimal places.

a $\log_8 21$

b $\log_4 \sqrt{5}$

c $\log_2 3^3$

d $\log_\pi 105$

17 If $p = \log_c 8$ and $q = \log_c 10$, write the following in terms of p and/or q:

a $\log_c 80$

b $\log_c \left(\frac{4}{5}\right)$

c $\log_c 100$

d $\log_c 64c$